

# **Mapping Africa's Endangered Archaeological Sites and Monuments (MAEASaM) Chronology Resource Model Documentation**

Arches Resource Model Graph and Reference Data (Thesaurus)

**Version 0.2 (current)**

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## Table of Contents

|                                                              |    |
|--------------------------------------------------------------|----|
| Section 1 Overview.....                                      | 1  |
| 1.1. Objectives and Scope.....                               | 1  |
| 1.2. Creation of the MAEASaM Chronology Resource Model ..... | 2  |
| 2. Section 2 Resource Model Structure.....                   | 2  |
| 2.1. Chronology graph structure.....                         | 2  |
| 2.2. Grid Resource Model Nodes and CIDOC-CRM Structure ..... | 4  |
| Semantic Node: Identifier .....                              | 5  |
| Semantic Node: Chronology Name.....                          | 5  |
| Semantic Node: Chronology Type.....                          | 7  |
| Semantic Node: Geographic Scope.....                         | 8  |
| Semantic Node: Time span .....                               | 9  |
| Semantic node: Description .....                             | 14 |
| Semantic node: Additional Chronology information .....       | 15 |
| Semantic node: Source.....                                   | 16 |
| Semantic node: Relationship to other temporal markers.....   | 17 |
| Semantic node: Resource version control .....                | 19 |
| 3. Appendix I .....                                          | 22 |

## Document history

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## Section I Overview

The MAEASaM Chronology Resource Model is a collection of linked African archaeological periods and other fine-grained temporal markers based on existing authoritative sources. An important component of the project's chronology resource model is that periods are considered space-time volumes; they contain a geographical extent with a temporal extent.

Each chronology record is fixed with a unique identifier that other resource models in the MAEASaM Arches database can refer to. The Chronology Resource Model is a support model that connects to two primary models in the database, Site and Remote Sensing. The model currently contains a 533-strong thesaurus of concepts relating to timespans in African archaeology that are currently managed in the Reference Data Manager of the project Arches database.

### I.1. Objectives and Scope

The rationale for the creation of a separate Chronology Resource Model was to address some of the complexities around defining the geographical and temporal extents of chronological periods which may use the same or similar names but which have been differently conceptualised depending on the region of study. This would also allow us to cater for the diversity of time-related concepts used within African archaeology and heritage studies as well as to relate such terms more precisely, and where possible, to the metadata associated with known temporal and spatial information.

Three questions guide the structure of the Resource Model:

- What is the temporal extent of a particular period?
- What is the geographical extent of a particular period?
- How do different periods relate to one another?

Similar to [iDAI ChronOntology](#) and [PeriodO](#), the creation of the Chronology Resource does not aim to resolve dating problems or make decisions on what constitutes a good dating classification-system. The resources collected are based on authoritative sources such as peer-reviewed scholarly articles and where necessary consultation with professional archaeologists involved in research of a particular period within a given region.

**Geographical extent:** as cultural periods are not confined to present-day political boundaries, the model is developed to include three broad geographical regions of Africa (West Africa (15 countries), Eastern Africa (14 countries), and Southern Africa (10 countries)). Whilst the MAEASaM Chronology resources are focused on eleven African countries it may also include neighbouring countries where broader, overlapping, time-space concepts have been defined.

**Temporal extent:** the MAEASaM project focuses on the documentation of sites of archaeological significance. This means that the timeline begins with the first appearance of material cultural assemblages. This is established by the first appearance of lithic industries belonging to the Oldowan of the Early Stone Age, dating to around 2.7 million years ago. The decision to mark the end of the

chronological timescale within the resource model however is more complex given that there exists several differences in how national heritage legislations define heritage. This means that all known periods up and into the ‘contemporary period’ have been captured.

## 1.2. Creation of the MAEASaM Chronology Resource Model

The Resource Model is designed using the conceptual modelling approach informed by the CIDOC Conceptual Reference Model (CIDOC-CRM) and its extension with the CRMgeo class SP12 Spacetime Volume Expression. It is implemented within the Arches platform as a graph-based data structure composed of nodes and relationships.

An important consideration made in the design of the resource model was that it had to be flexible enough to allow for several time spans (some general and some quite specific). It also needed to account for ‘fuzziness’ or uncertainty in time ranges (dates) and/or spatial extents.

The development of the chronology records has benefitted from the consultation of already well-established public domain chronology databases, particularly **PeriodO** (<https://perio.do/en/>), a gazetteer of periods for linking and visualising data, **iDAI ChronOntology** (<https://chronontology.dainst.org/>), a webservice developed by the German Archaeological Institute, that connects period terms to datings and the online **Art & Architecture Thesaurus from the Getty Research Institute (Getty AAT)**.

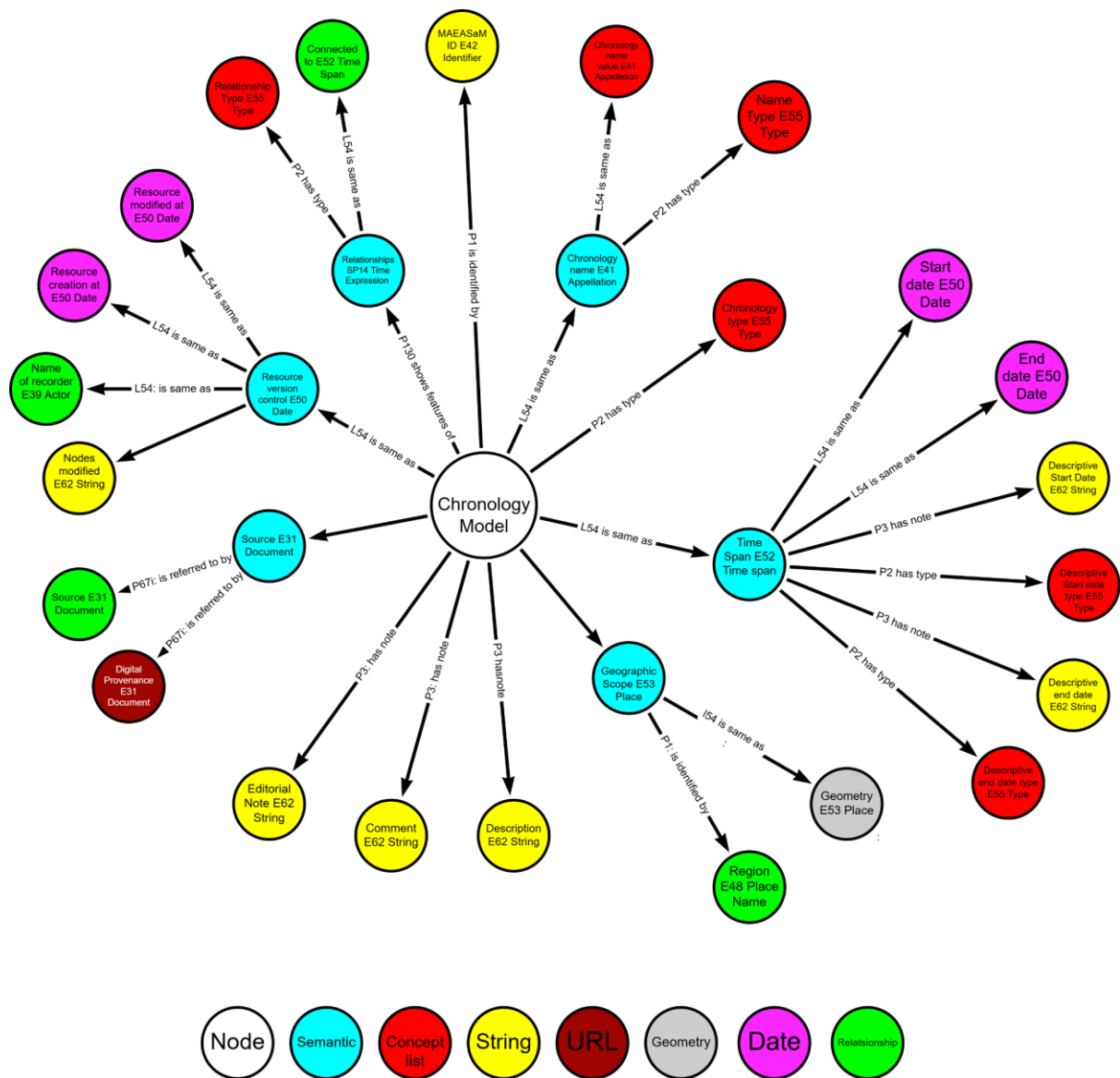
A comprehensive set of resources covering African chronological concepts has been collected which expands on pre-existing African archaeological databases.

## 2. Section 2 Resource Model Structure

### 2.1. Chronology graph structure

The Chronology Resource Model is implemented as a graph-based structure in which nodes represent specific types of information and relationships that represent logical connections between them.

The model includes information pertaining to the name of the resource, its start and end dates, a general description followed by its geographical location and additional original source information.



Structure of the Chronology Graph Model

## 2.2. Grid Resource Model Nodes and CIDOC-CRM Structure

### Schema Attributes and Definitions

**Group Name:** In Arches, this corresponds to the Semantic Node. It represents the parent category under which related metadata elements are organised.

**Node Name:** Also referred to as a metadata element. This is the primary descriptive field used to document a Remote Sensing record.

**Node Function:** Provides a functional description of the node, explaining its role within the resource model.

**CIDOC CRM Class:** Indicates the equivalent class in the CIDOC CRM ontology. This mapping supports semantic interoperability.

**CIDOC CRM Property:** Indicates the equivalent property in CIDOC CRM ontology. This mapping supports semantic interoperability.

**Connector Node:** Specifies whether the node establishes a relationship with another resource model. Values are “yes” or “no”.

**Connected To:** If the Connector Node value is “yes”, this field specifies the name of the related Resource Model.

**Data Type:** Defines the type of data entered in Arches. Common data types include: string, date, concept list, and resource-instance-list.

**Necessity:** Indicates whether the node is required when describing a record. Possible values are:

“mandatory”

“Strongly suggested”

**Cardinality:** Specifies how many values a node can contain. In the MAEASaM Arches context, four options are used:

1 - Only one value allowed

0..1 - Optional, maximum one value

1.. n - At least one value required, multiple allowed

0.. n - Optional, multiple values allowed

**Scope Note:** Provides a definition and conceptual explanation of the node.

**Protocol for Entry:** Explains how information should be entered into this node, including formatting rules, conventions, or controlled vocabulary guidance.

### Semantic Node: Identifier

**Node Name:** MAEASaM ID

**Node Function:** Identifier

**CIDOC CRM Class:** E42 Identifier

**CIDOC CRM Property:** P1 is identified by

**Connector Node:** No

**Connected to:** None

**Node type:** String

**Necessity:** Mandatory

**Cardinality:** 1

#### Scope note (description of node):

A unique ID is given to each resource entered. Every resource must have exactly one unique MAEASaM ID.

#### Protocol for entry:

Each chronology record in the database is assigned a unique ID composed of two parts.

The first part shows the type of record and, in the Chronology Model, is always written as CHRN, which stands for Chronology. The second part is a number that indicates the order in which the chronology resource was entered. The number is always written with eight digits with zeros added at the beginning. For example: 00000001, 00000010, 00000100).

A resource cannot be created without this identifier, and no resource may have more than one MAEASaM ID.

#### Example:

CHRN-00000001

CHRN-00000010

CHRN-00000100

### Semantic Node: Chronology Name

**Node Name:** Chronology name value

**Node Function:** Appellation

**CIDOC CRM Class:** E41 Appellation

**CIDOC CRM Property:** L54 is same as

**Connector Node:** No

**Connected to:** None

**Node type:** Concept list (Chronology name value)

**Necessity:** Mandatory

**Cardinality:** 1..n

**Scope note (description of node):**

This node captures the name of the given chronology record. A term may be given a qualifier to distinguish it from a similar or same term but with different meanings or spatial context. Where applicable, qualifiers have been represented by parentheses ( ) at the end of a term.

**Protocol for entry:**

This node should be used to record the name or term used to describe the chronology concept.

- The node is mandatory, and only one term may be selected from the concept list
- Where appropriate, select more specific terms to describe the chronology entry.
- The use of qualifiers at the end of a term must include a parenthesis ( ) containing a term to distinguish it from another similar or same concept.

**Examples of Chronology name value:**

Later Stone Age (Eastern Africa)

Bambata industry (MSA)

Early Aksumite phase

**Examples of Chronology name value qualifiers:**

Moloko (ceramic)

Ghana empire (polity)

Great Zimbabwe type walling (architecture)

Stone Age (West Africa)

Earlier Stone Age (Senegal)

Earlier Stone Age (Mali)

**Node Name:** Chronology name type

**Node Function:** Type assignment

**CIDOC CRM Class:** E55 Type

**CIDOC CRM Property:** P2 has type

**Connector Node:** No

**Connected to:** None

**Node type:** Concept list (Name type)

**Necessity:** Mandatory

**Cardinality:** 1..n



**Scope note (description of node):**

Chronology name type defines the name of the chronology resource entry as either primary name, an alternative name or a local name, etc. and derives from a concept list.

**Protocol for entry:**

- Entry is mandatory.
- Alternative terms may be added and must be specified

**Examples:**

Later Stone Age (Eastern Africa) = Primary name

LSA (Eastern Africa) = Alternative Name

**Semantic Node: Chronology Type**

**Node Name:** Chronology name type

**Node Function:** Type assignment

**CIDOC CRM Class:** E55 Type

**CIDOC CRM Property:** P2 has type

**Connector Node:** No

**Connected to:** None

**Node type:** Concept list (Chronology type)

**Necessity:** Mandatory

**Cardinality:** 1..n

Chronology Type is a node that captures the defining characteristic of a period or chronological temporal marker in simple terms. More than one chronology type can be selected for a chronology resource entry. In case the defining characteristic of the time period is not clear in the original source (i.e. not stated or undefined) the type “Non-specified” can be used. However, this is best avoided.

MAEASaM has compiled 8 broad chronology types each of which may contain subgroupings:

1. Geological Timescale
2. Archaeological Period
3. Architectural style
4. Local sequence
5. Material culture
6. Oral or documented marker
7. Polity
8. Palaeoclimate or palaeoenvironmental reconstruction

The description of each concept may be used as a guide for the general characteristics from which the chronology resource entry has been determined. Multiple chronology types can be used.

**Protocol for entry:**

- This is a concept-list field that requires at least one concept to be selected. Multiple concepts may be chosen where relevant.
- Based on the concept description, identify the primary classification system used to define the period. For example, *Moloko* is classified mainly based on a specific ceramic facies that has been used to define a temporal period in southern African archaeology.
- If no specific classification system is referenced, select “**Non-specified.**”

**Examples:**

Geological epoch  
Ceramic tradition  
Ceramic facies  
Lithic industry

**Semantic Node: Geographic Scope**

**Node Name:** Region

**Node Function:** Relationship

**CIDOC CRM Class:** E48 Place Name

**CIDOC CRM Property:** PI\_is identified by

**Connector Node:** Yes

**Connected to:** Administrative Resource Model

**Node type:** Resource instance list

**Necessity:** Strongly recommended

**Cardinality:** 0..n

**Scope note (description of node):**

This node is connected to the Administrative Resource Model and is identified by a place name (Administrative area) where the geographical extent of the chronology resource covers. More than one relationship may be made.

**Protocol for entry:**

- Based on the geographical extent of the chronology resource, identify the corresponding administrative areas that the resource connects to.
- More than one administrative area may be selected
- Where possible, be specific about the area of coverage.
- Use the unique MAEASaM ID derived from the Administrative Resource to enter into the Chronology metadata template.

**Example:**

| Example of Region | Example of unique MAEASaM ID to enter into Chronology Metadata template |
|-------------------|-------------------------------------------------------------------------|
| Eastern Africa    | ADMIN-000000004                                                         |
| Ethiopia          | ADMIN-ETH-000000001                                                     |

**Node Name:** Geometry (WKT)

**Node Function:** geometry recording

**CIDOC CRM Class:** E53 Place

**CIDOC CRM Property:** L54\_is\_same\_as

**Connector Node:** None

**Connected to:** None

**Node type:** GeoJSON-feature-collection

**Necessity:** Strongly recommended

**Cardinality:** 1

**Scope note (description of node):**

The geometry recorded in this node represents the best available spatial approximation of the chronology at the time of recording. It may take the form of a polygon or multi-polygon, depending on the nature of the distribution of chronology resources and the available data.

The geometry is recorded in Well Known Text (WKT) format and converted to GeoJSON-feature-collection when uploaded to Arches database.

**Protocol for entry:**

The function of this node is to record the geographic location and spatial extent of the identified

- Enter only one WKT (either polygon or multipolygon) per MAEASaM ID
- Ensure that the entry of the WKT contains a X,Y & Z value
- More precise geometries may be added or updated as improved data becomes available

Chronology geometries are captured as polygons or multipolygons in WKT

**Example:**

POLYGON ((30.82 21.35, 30.83 21.35, 30.83 21.36, 30.82 21.36, 30.82 21.35))

**Semantic Node: Time span**

A supplementary guideline on the entering of dates for MAEASaM Chronology resources is available in Appendix I of this guideline.

**Node Name:** Start date

**Node Function:** time

**CIDOC CRM Class:** E50 Date

**CIDOC CRM Property:** L54\_is\_same\_as

**Connector Node:** None

**Connected to:** None

**Node type:** Date

**Necessity:** Mandatory

**Cardinality:** 1

**Scope note (description of node):**

The node records the earliest known occurrence of the chronology period or 'start of existence' and is reported using the Extended Date/Time format (EDTF),

<https://www.loc.gov/standards/datetime/> .

The start date can be recorded with different levels of precision, depending on the information available. This may include an exact date, an approximate date, an uncertain date or a broader time range. Follow the EDTF guidelines for date entry [here](#).

Table 1: Summary adapted after the Extended Date Time Format (EDTF) Specification (Library of Congress) (loc.gov)

| Problem or uncertainty                                      | EDTF Modifier | Rules                                                                                                                       | Example of conversion               |
|-------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| When a date exceeds four digits.                            | Y             | 'Y' represents 'Year' in EDTF and should only be used when the date exceeds four digits.                                    | 10 000 BP >> Y-10000                |
| When a date is before common era (bce)                      | -             | A negative symbol should be used in front of a date if it is <i>Before Common Era</i> (BCE).                                | 150 BCE >> -150                     |
| When a date appears to be uncertain                         | ?             | The character (also referred to as a modifier) may be used only at the end of the date string and apply to the entire date. | Uncertain age of 1850 BCE >> -1850? |
| A date that is presented as a range with ( - )              | /             | Replace the hyphen ( - ) with a backslash (/)                                                                               | 800 BCE - 200 CE >> -800/200        |
| A date that is presented as 'approximately' or 'circa' (c.) | ~             | Convert strings to add tilde (~)                                                                                            | Circa 1850 CE >> 1850~              |
| A period with an unknown start or end date                  | ..            | Convert strings to add a double dot trailing the end of date.                                                               | unknown >> ..                       |
| 'on and before' Or 'on and after' a                         | {..}          | Convert strings to add curly brackets plus double dots                                                                      | All years before 1999 >> {...1999}  |

|      |  |  |  |
|------|--|--|--|
| date |  |  |  |
|------|--|--|--|

**Protocol for entry:**

- Where multiple start dates are reported for the same period in the same geographic region, then the earliest start date of existence should be selected.
- There should be no spaces between digits and modifiers in the EDTF reporting for start dates
- Where possible dates entered for the start of existence of a particular chronological period should be represented as calendrical years such as CE (Common Era) or BCE (Before Common Era). This will ensure consistency in the reading of the EDTF and for searchability.
- When entering dates beyond the extent of the radiocarbon curve and beyond the calendrical dating system, then before present (bp) can be used. This will be noted in the 'Descriptive Date Type' node.

**Examples:**

| Chronology name value     | Start date using EDTF |
|---------------------------|-----------------------|
| 2nd Millennium BCE        | -2000                 |
| 1st Millennium CE         | 1                     |
| Iron Age (Eastern Africa) | -2600                 |

**Node Name:** End date**Node Function:** time**CIDOC CRM Class:** E50 Date**CIDOC CRM Property:** L54\_is\_same\_as**Connector Node:** None**Connected to:** None**Node type:** Date**Necessity:** Mandatory**Cardinality:** 1**Scope note (description of node):**

The node records the latest known occurrence of the chronology period or 'end of existence' and is reported using the Extended Date/Time format (EDTF),

<https://www.loc.gov/standards/datetime/> .

Like the Start date, the end date can be recorded with different levels of precision, depending on the information available. This may include an exact date, an approximate date, an uncertain date or a broader time range. Follow the EDTF guidelines for date entry [here](#) (refer to Table I, pg. 13).

**Protocol for entry:**

- Where multiple end dates are reported for the same period in the same geographic region, then the latest end date of existence should be selected.
- There should be no spaces between digits and modifiers in the EDTF reporting for end dates
- Where possible dates entered for the end of existence of a particular chronological period should be represented as calendrical years such as CE (Common Era) or BCE (Before Common Era). This will ensure consistency in the reading of the EDTF and for searchability.
- When entering dates beyond the extent of the radiocarbon curve and beyond the calendrical dating system, then before present (bp) can be used. This will be noted in the 'Descriptive Date Type' node.

**Examples:**

| Chronology name value     | End date using EDTF |
|---------------------------|---------------------|
| 2nd Millennium BCE        | -1001               |
| 1st Millennium CE         | 1000                |
| Iron Age (Eastern Africa) | 1600                |

**Node Name:** Descriptive start date

**Node Function:** free text

**CIDOC CRM Class:** E62\_String

**CIDOC CRM Property:** P3\_has note

**Connector Node:** None

**Connected to:** None

**Node type:** String

**Necessity:** Mandatory

**Cardinality:** 1

**Scope note (description of node):**

The descriptive start date reports the original start of existence date as described from the authoritative source before it has been converted to the Extended Date/Time Format (EDTF).

**Protocol for entry:**

- Where multiple start dates are reported for the same period in the same geographic region, then the earliest of the start of existence should be selected.
- The descriptive start date should be completed along with the 'Descriptive start date type'
- Only one entry is required for this node.

**Example:**

956

**Node Name:** Descriptive start date type

**Node Function:** identifier

**CIDOC CRM Class:** E55\_Type

**CIDOC CRM Property:** P2\_has type

**Connector Node:** None

**Connected to:** None

**Node type:** Concept list (Descriptive start and end date)

**Necessity:** Mandatory

**Cardinality:** 1

**Scope note (description of node):**

This node uses a concept list to describe the calendrical system where the original start date was reported (i.e., deriving from an authoritative source).

**Protocol for entry:**

- This node makes use of the Descriptive start and end date concept list with no more than one entry being permitted.
- Where information about the calendrical system used to describe the date is missing, '**Non-specified**' may be used, however this is best avoided.

**Example:**

bce - before common era

ce - common era

BP - before present

**Node Name:** Descriptive end date

**Node Function:** free text

**CIDOC CRM Class:** E62\_String

**CIDOC CRM Property:** P3\_has note

**Connector Node:** None

**Connected to:** None

**Node type:** String

**Necessity:** Mandatory

**Cardinality:** 1

**Scope note (description of node):**

The descriptive end date reports the original end of existence date as described from the authoritative source before it has been converted to the Extended Date/Time Format (EDTF).

**Protocol for entry:**

- Where multiple end dates are reported for the same period in the same geographic region, then the latest of the end of existence should be selected.
- The descriptive end date should be completed along with the 'Descriptive end date type'
- Only one entry is required for this node.

**Example:**

1640

**Node Name:** Descriptive end date type

**Node Function:** identifier

**CIDOC CRM Class:** E55\_Type

**CIDOC CRM Property:** P2\_has type

**Connector Node:** None

**Connected to:** None

**Node type:** Concept list (Descriptive start and end date)

**Necessity:** Mandatory

**Cardinality:** 1

**Scope note (description of node):**

This node uses a concept list to describe the calendrical system where the original end date was reported (i.e., deriving from an authoritative source).

**Protocol for entry:**

- This node makes use of the Descriptive start and end date concept list with no more than one entry being permitted.
- Where information about the calendrical system used to describe the date is missing, '**Non-specified**' may be used, however this is best avoided.

**Example:**

bce - before common era

ce - common era

BP - before present

**Semantic node: Description**

**Node Name:** Description

**Node Function:** description

**CIDOC CRM Class:** E62\_String

**CIDOC CRM Property:** P3\_has note

**Connector Node:** None

**Connected to:** None



**Node type:** String  
**Necessity:** Mandatory  
**Cardinality:** 1..n

**Scope note (description of node):**

This node includes the definition and/or defining characteristics of the period or temporal marker.

**Protocol for entry:**

- This is a free text node that includes definitions and/or defining characteristics described for the period or temporal marker as based on an authoritative source. It is important to be as detailed as possible, and include information on geographic and temporal extent within the text.
- Where statements have been directly extracted from the original source then the author name, date of publication and page number, contained within parentheses, are to be provided at the end of the definition. Avoid the use of quotation marks when providing a definition.
- This is a mandatory node where more than one definition may be provided.

**Example:**

Diamant facies is a type of pottery characterised by tapered rims with broadly incised herringbone decoration. It is found in southeastern Botswana and in northwestern South Africa. Most likely date range: CE 750 - CE 1000.

**Semantic node: Additional Chronology information**

**Node Name:** Comment  
**Node Function:** free text  
**CIDOC CRM Class:** E62\_String  
**CIDOC CRM Property:** P3\_has note  
**Connector Node:** None  
**Connected to:** None  
**Node type:** String  
**Necessity:** Strongly recommended  
**Cardinality:** 0..n

**Scope note (description of node):**

The node captures additional information not captured in the description or definitions. Such comments may include but are not limited to debates, geographical coverage and/or dating issues.

**Protocol for entry:**

- This free-text field is used to flag relevant past and current scholarly debates associated with the recorded period and/or temporal marker.

- It may also include additional contextual information related to uncertainties or discussions about dating and/or geographic extent.
- References should be cited using the original author's surname and year of publication, enclosed in parentheses.

**Example:**

Bambata A is suggested to belong to the Later Stone Age herders of southern Africa and Bambata B with farming communities. The connection between Bambata wares and the larger Kalundu ceramic tradition is debated (Walker 1983, Sadr 2003, Huffman 2007, 2021).

**Node Name:** Editorial note

**Node Function:** free text

**CIDOC CRM Class:** E62\_String

**CIDOC CRM Property:** P3\_has note

**Connector Node:** None

**Connected to:** None

**Node type:** String

**Necessity:** Strongly recommended

**Cardinality:** 0..n

**Scope note (description of node):**

This node records any changes made by the recorder to the definition and/or geometry, and/or start and end dates for the resource entry.

**Protocol for entry:**

- This is a free text field that allows the recorder to capture any changes and/or modifications made to the definition, geometry, and/or start and end dates as part of the data entry process.

**Example:**

Description adapted after John Bower (1991) with geometry modified after Grillo et al. (2018).

**Semantic node: Source**

**Node Name:** Source

**Node Function:** relationship

**CIDOC CRM Class:** E31\_Document

**CIDOC CRM Property:** P129i\_is subject\_of

**Connector Node:** Yes

**Connected to:** Information Resource Model

**Node type:** Resource instance list

**Necessity:** Mandatory

**Cardinality:** 1..n

**Scope note (description of node):**

The node is a resource instance list meaning that it is connected to information deriving from the Information Resource Model. The Information Resource Model contains important metadata (e.g., url links, author and publication details) linked to the original authoritative source from where the chronology record has been derived.

**Protocol for entry:**

- Identify the correct metadata relating to the original source information for the chronology record and enter only the corresponding unique MAEASaM ID for the specified information resource.
- More than one information resource may be linked to the chronology record, using a separate row for each related information record, complete by filling in the corresponding unique MAEASaM ID.

**Example:**

INFO-00000651

INFO-00000697

**Node Name:** Digital provenance

**Node Function:** permalink

**CIDOC CRM Class:** E31\_Document

**CIDOC CRM Property:** P165i\_is incorporated\_in

**Connector Node:** None

**Connected to:** None

**Node type:** ?

**Necessity:** Strongly recommended

**Cardinality:** 0..1

**Scope note (description of node):**

The node refers to the permalink from where the original term derives and uses only a url link to connect the record to the original database.

**Protocol for entry:**

- This information should only be entered when an online thesaurus (e.g. Getty AAT, ChronOntology, PeriodO) is used to create the resource.
- Use this node to record the permalink from which the original term is derived.

**Example:**

<http://n2t.net/ark:/99152/p0s2rwk>

**Semantic node: Relationship to other temporal markers**

**Node Name:** Relationship type

**Node Function:** identifier

**CIDOC CRM Class:** E55\_Type

**CIDOC CRM Property:** P2\_has type

**Connector Node:** None

**Connected to:** None

**Node type:** Concept list (Relationship type)

**Necessity:** Strongly recommended

**Cardinality:** 0..1

**Scope note (description of node):**

Relationship type specifies in basic terms the type of relationship between two chronology resources. The node uses a concept-list.

**Protocol for entry:**

- Using the concept list, select the correct relationship type (one entry only) that best describes the relationship between two chronology records.
- Relationships may be complex or simple with multiple being made using different rows in the Chronology metadata template, each of which corresponding to different chronology records.

**Example:**

Stone Age (Southern Africa) **has broader term (has narrower term)** Stone Age (Southern Africa)

**Node Name:** Connected to

**Node Function:** relationship

**CIDOC CRM Class:**

**CIDOC CRM Property:**

**Connector Node:** Yes

**Connected to:** Chronology Resource Model

**Node type:** Resource instance list

**Necessity:** Strongly recommended

**Cardinality:** 0..1

**Scope note (description of node):**

Like the Region and Source fields, the 'Connected to' node refers to a resource instance list connecting one period or temporal marker to another. The unique MAEASaM ID for the connected Chronology record is to be entered.

**Protocol for entry:**

- Identify the associated or related chronology record to the record you are currently working on.
- Enter the unique MAEASaM ID for the connected chronology record into the specified field.
- Note that more than one record may be connected, however, these will be considered as separate entries and will therefore require separate rows in the Chronology metadata template for input, alongside a specified relationship type field.

**Example:**

**CHRN-00000456** has broader term (has narrower term) **CHRN-00000451**

#### Semantic node: Resource version control

**Node Name:** Resource creation at

**Node Function:** Date

**CIDOC CRM Class:** E63\_Beginning of existence

**CIDOC CRM Property:** P92i\_was\_brought\_into\_existence\_by

**Connector Node:** None

**Connected to:** None

**Node type:** Date

**Necessity:** Mandatory

**Cardinality:** 1

#### Scope note (description of node):

The date at which the chronological record was entered into the MAEASaM Arches database.

It documents the initial creation timestamp of the resource entry and forms part of the version control mechanism of the system.

Dates may vary in their degree of precision.

#### Protocol for entry:

- This field is automatically generated by the system at the time the resource is created.
- Users must not manually edit this value.
- The date should follow the system's standard date format.

#### Example:

08-04-2026

**Node Name:** Resource modified at

**Node Function:** Date

**CIDOC CRM Class:** E50\_Date

**CIDOC CRM Property:** L54\_is\_same\_as

**Connector Node:** None

**Connected to:** None

**Node type:** Date

**Necessity:** Mandatory

**Cardinality:** 1

#### Scope note (description of node):

The date at which the chronological record has been modified in the MAEASaM Arches database.

It supports tracking updates and changes to the resource over time.

**Protocol for entry:**

- This field is automatically updated by the system whenever the resource is edited.
- Users must not manually edit this value.

**Example:**

20-04-2026

**Node Name:** Name of recorder

**Node Function:** Actor

**CIDOC CRM Class:** E39 Actor

**CIDOC CRM Property:** P1 is identified by

**Connector Node:** Yes

**Connected to:** Actor Resource Model

**Node type:** Resource instance list

**Necessity:** Mandatory

**Cardinality:** 1

**Scope note (description of node):**

This node records the name of the individual responsible for creating the resource record.

**Protocol for entry:**

- The recorder must be selected from the existing Actor records in the database.
- Free text entries should not be used when a corresponding Actor record exists.
- This field is mandatory and must be completed at the time of resource creation.

**Example:**

ACT-00000047

**Node Name:** Nodes modified

**Node Function:** free text

**CIDOC CRM Class:** E62 String

**CIDOC CRM Property:** P3 has note

**Connector Node:** None

**Connected to:** None

**Node type:** String

**Necessity:** Strongly recommended

**Cardinality:** 0..n

**Scope note (description of node):**

This node records the names of the nodes that were modified during an update of the resource record.

It supports internal version tracking and documentation of changes.

**Protocol for entry:**

- Using free text to complete the field, specify the names of the nodes (fields) that were modified by the recorder.
- Multiple entries may be made.

**Example:**

Node name: Description

## 3. Appendix I

### MAEASaM Chronology Time Span

#### Section 1: MAEASaM Chronology Time Span

- 1.1 MAEASaM Timespan
- 1.2 Start Date and End Date: branches of Timespan
- 1.3 Scopenote
- 1.4 Standardization of time
- 1.5 Extended Date/Time Format (EDTF)
- 1.6 Recommended protocols for EDTF entry
- 1.7 Recommended protocols for conversions to EDTF

#### Section 2: MAEASaM Chronology Descriptive Start and End Dates

- 2.1 Reporting original date derived from source
- 2.2 Descriptive date
- 2.3 Recommendations for entry

#### Section 3: MAEASaM Chronology Comments and Editorial Note for Start and End Dates

- 3.1 MAESAM Comment
- 3.2 Comment Scope note
- 3.3 Recommendations for entry in connection to reporting of dates
- 3.4 MAEASaM Editorial Note
- 3.5 Editorial Note Scope Note
- 3.6 Recommendations for entry in connection to reporting of dates

#### Appendix I: Recommendations for original date conversions

- 1.1 Recommendations for reporting centuries or millennia
- 1.2 Radiocarbon age determinations (bp)
- 1.3 Recommendations for radiocarbon calibration
- 1.4 Recommendations for Calibration Softwares
- 1.5 Method for calibration using Calib



This document details the protocols, guidelines, and standards used to treat start and end dates in the Time Span (CIDOC-CRM E52) branch of the MAEASaM Chronology Resource Model. It includes the use of the Extended Date/Time Format (EDTF) in the translation of original recorded dates from publications on African archaeology. This document summarizes the protocols used with the treatment of absolute dates including calibration procedures and the use of the ‘editorial note’ to annotate important contextual information that may not be captured in related fields of the Chronology Resource Model.

## I.1 MAEASaM Time Span

Node Name: **Time span**

Node Function: **start of existence and end of existence of a particular period**

CIDOC CRM Class: **E52**

The screenshot displays the front-end user interface for creating a new Time Span resource. On the left, a sidebar contains a tree view of the resource model hierarchy. The 'Time span' node is selected and highlighted. The main content area is titled 'New Time span' and features two input fields for 'Start date' and 'End date'. Each field has a dropdown menu labeled 'EDTF Formats' to the right. Below these fields is a green button with a plus icon and the text 'Add'.

**Figure 1.** Front end user interface of MAEASaM Chronology Time Span with start and end dates represented as Extended Date/Time Format (EDTF)

## I.2 Start Date and End Date: branches of Timespan

Node Name: **Start Date**  
 Node Function: **Date**  
 CIDOC CRM Class: **E50**

Node Name: **End Date**  
 Node Function: **Date**  
 CIDOC CRM Class: **E50**

### I.3 Scope note

Start date refers to the lower value or the earliest of the period of existence (i.e. when a particular chronology begins). End date refers to the higher value or the most recent of the period of existence (i.e. when a particular chronology ceases to exist). MAEASaM Chronology uses the Extended Date/Time Format (EDTF) conformance levels 0, 1 and 2 to indicate start and end dates. The Start and End Dates described under this section relate specifically to the EDTF system and are different to 'Descriptive Dates' as discussed in Section 2 of this document.

### I.4 Standardization of time

The Extended Date/Time Format (EDTF) was created by the Library of Congress with the participation and support of the bibliographic community as well as communities with related interests. It defines features to be supported in a date/time string, features considered useful for a wide variety of applications. Three conformance levels are defined: level 0, level 1, and level 2. Level 0 specifies features of ISO 8601-1; Levels 1 and 2 specify features described ISO 8601-2). For more information about Extended Date/Time Format [Extended Date Time Format \(EDTF\) Specification \(Library of Congress\) \(loc.gov\)](https://www.loc.gov/edtf/).

Extended Date/Time Format is a profile of ISO\_8601 and its extension ISO 8601-2:2019. EDTF includes qualifiers to indicate uncertainty in dates and time intervals with an open end or unknown end. For more information about ISO\_8601 and ISO 8601-2:2019 [https://en.wikipedia.org/wiki/ISO\\_8601#Calendar\\_dates](https://en.wikipedia.org/wiki/ISO_8601#Calendar_dates)

### I.5 Extended Date/Time Format (EDTF)

Table I below summarizes the format for entering start and end dates using the Extended Date/Time Format (EDTF) for the MAEASaM Chronology Resource Model. Visit the official website [Extended Date Time Format \(EDTF\) Specification \(Library of Congress\) \(loc.gov\)](https://www.loc.gov/edtf/) for further details.

**Table I:** Table summary adapted after: [Extended Date Time Format \(EDTF\) Specification \(Library of Congress\) \(loc.gov\)](https://www.loc.gov/edtf/)

| Problem or uncertainty | EDTF Modifier | Rules | Example of conversion |
|------------------------|---------------|-------|-----------------------|
|------------------------|---------------|-------|-----------------------|

|                                                             |      |                                                                                                                             |                                     |
|-------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| When a date exceeds four digits.                            | Y    | 'Y' represents 'Year' in EDTF and should only be used when the date exceeds four digits.                                    | 10 000 BP >> Y-10000                |
| When a date is before common era (bce)                      | -    | A negative symbol should be used in front of a date if it is <i>Before Common Era</i> (BCE).                                | 150 BCE >> -150                     |
| When a date appears to be uncertain                         | ?    | The character (also referred to as a modifier) may be used only at the end of the date string and apply to the entire date. | Uncertain age of 1850 BCE >> -1850? |
| A date that is presented as a range with ( - )              | /    | Replace the hyphen ( - ) with a backslash (/)                                                                               | 800 BCE - 200 CE >> -800/200        |
| A date that is presented as 'approximately' or 'circa' (c.) | ~    | Convert strings to add tilde (~)                                                                                            | Circa 1850 CE >> 1850~              |
| A period with an unknown start or end date                  | ..   | Convert strings to add a double dot trailing the end of date.                                                               | unknown >> ..                       |
| 'on and before' Or 'on and after' a date                    | {..} | Convert strings to add curly brackets plus double dots                                                                      | All years before 1999 >> {..1999}   |

## I.6 Recommended protocols for Extended Date/Time Format entry

- There should be no spaces between digits and modifiers in the EDTF reporting for Start and End Dates.

## I.7 Recommendations for conversions to Extended Date/Time Format

- Where possible dates entered for start and end of existence should be represented as calendrical years such as CE (Common Era) or BCE (Before Common Era). This will ensure consistency in the reading of the EDTF and for searchability. Refer to Appendix I for recommended protocols for the numerical representation of centuries and millennia as used in the MAEASaM Chronology (pp. X).
- Where radiocarbon dates (bp) are provided they must be converted to calendrical years using a process called calibration. Refer to Appendix I for recommended protocols for the calibration of radiocarbon dates as used in the MAEASaM Chronology (pp. X). A note of the radiocarbon curve, correction of the radiocarbon reservoir effect, and probability range, type of material dated, the radiocarbon laboratory ID, and where possible, the contextual information, should be noted in the 'Editorial Note' field of the Chronology Resource Model. For further details on calibration protocols refer to Section 3: MAEASaM Chronology Comment and Editorial Note for Start and End Dates (pp. X)

- c. If you are working with ages beyond the extent of the radiocarbon curve then before present (bp) dates can be used. This will be noted in the 'Descriptive Date Type' field of the Timespan branch of the Chronology Resource Model.
- d. Where dates are reported without indication of whether they have been calibrated or not and/or are reported as Before Present (BP), this should be noted in the 'Descriptive Date Type' field of the Timespan branch of the Chronology Resource Model. Additional information pertaining to any problematic date can be annotated in the 'Editorial Note' field of the Chronology Resource Model.
- e. Where absolute dates have been modeled using Bayesian methods or any other, a note of this should be made in the 'Editorial Note' field of the Chronology Resource Model. The raw radiocarbon date is preferred over modeled dates in the chronology resource.

## Section 2: MAEASaM Chronology Descriptive Start and End Dates

New node to be added to the Chronology Resource Model: Descriptive Date. This will allow us to indicate original date reporting as CE, BCE, BP, and/or ranges.

Option 1:

Start date: -2000

Descriptive date (or any name you like): 2000

Descriptive Type: BCE (edited)

Option 2:

Start date: -2000

Descriptive date (or any name you like): 2000 BCE

Note: With option 2, there will be no control of how users will enter BCE (for example, they could enter Before Common Era).

## Section 3: MAEASaM Chronology Comment and Editorial Note for Start and End Dates

This section covers recommendations for the use of the 'Editorial Note' and 'Comment' section in connection with Start and End dates of the MAEASaM Chronology'.

The editorial note is a string (free text) node which allows users to make use of notes for editorial purposes. This node is a branch of the 'Comment' section of the Chronology also referred to as 'Annotation Object' (E29) in CIDOC-CRM ontology. The descriptions of 'Comment' and Editorial Note in the Chronology are detailed below.

### 3.1 Comment

### 3.3 Recommendations for entry in connection to the reporting of dates

- Specify if start and end dates are debated and provide author name and date. Full sources will be entered into the Source Field.
- Specify if dates are problematic and provide author name and date. Full sources will be entered into the Source Field.
- Specify if alternative dates have been suggested for the entry, provide alternative date(s) and author name and date. Full sources will be entered into the Source Field.

### 3.4 Editorial Note

Node Name: **Editorial Note**

Node Function: **String (free text)**

CIDOC CRM Class: **E29 (Annotation Object)**

The screenshot displays the 'New Resource' interface for MAEASaM. On the left, a sidebar shows a tree view of the resource model, with 'Comment' selected under 'New Resource (Chronology)'. The main area is titled 'New Comment'. It features a 'Comment' section with a rich text editor (including bold, italic, underline, link, and other formatting tools) and a 'Source' field. Below this, an 'Editorial Note' section is highlighted with an orange border, containing a text input field labeled 'Enter text' and a green '+ Add' button.

Figure X: Front end user interface of MAEASaM Chronology Editorial Note

### 3.5 Scope note

A (free text) string node that allows notes to be made by the editor of the entry based on any modification that is made to the entry upon input into the database.

### 3.6 Recommendations for entry in connection to reporting of dates

**A. If a date has not been calibrated the following should be noted in the Editorial Note Section:**

- *Uncalibrated, the date is reported as original from source.*

**B. If a radiocarbon date is used the following should be noted in the Editorial Note section:**

- *Specify that the radiocarbon age that has been calibrated.*
- *Specify the radiocarbon curve correction (i.e. SHCal 20 or IntCal20 or Marine 20 (including the  $\Delta R$  value (i.e.  $71 \pm 13$  BP)) (See Appendix 1.2).*
- *Specify the probability range used (i.e. at 2 sigma or 95.4% probability). Note that the recommended probability range should be set at 2 sigma for calibration (98% probability).*
- *Specify the sample from which the date derives.*
- *Specify the Laboratory Number from which the date derives*
- *Example:*

*Editorial Note*

*Date calibrated, SHCal20 at 95% probability, charcoal sample, Lab ID PTA – 2225*

- If statistical methods are used on radiocarbon dates for chronological purposes (i.e. Bayesian modeling) then this should be noted in the Editorial Note of the Chronology.

**C. If a date status is unknown for example, it is reported as BP but does not provide additional information for the original radiocarbon age, the following should be noted in the Editorial Note:**

- *The date(s) is reported as original from source.*

## Appendix I

### Original date conversions for Extended Date Time Format (EDTF)

## I.1 Protocols for reporting centuries or millennia

To retain consistency in the reporting of centuries for chronological **start date** and **end date** entries, we use the following convention:

| <i>Original Date Reported</i> | <i>Start Date (EDTF)</i> | <i>End Date (EDTF)</i> |
|-------------------------------|--------------------------|------------------------|
| <i>Century (100 yrs)</i>      |                          |                        |
| 12th Century BCE              | -1199                    | -1100                  |
| 11th Century BCE              | -1099                    | -1000                  |
| 10th Century BCE              | -999                     | -900                   |
| 3rd Century BCE               | -299                     | -200                   |
| 2nd Century BCE               | -199                     | -100                   |
| 1st Century BCE               | -99                      | -1                     |
| 1st Century CE                | 1                        | 99                     |
| 2nd Century CE                | 100                      | 199                    |
| 3rd Century CE                | 200                      | 299                    |
| 10th Century CE               | 900                      | 999                    |
| 11th Century CE               | 1000                     | 1099                   |
| 12th Century CE               | 1100                     | 1199                   |
| 19th Century CE               | 1800                     | 1899                   |
| <i>Millennium (1000 yrs)</i>  |                          |                        |
| 4th Millennium BCE            | -3999                    | -3000                  |
| 3rd Millennium BCE            | -2999                    | -2000                  |
| 2nd Millennium BCE            | -1999                    | -1000                  |
| 1st Millennium BCE            | -999                     | -1                     |
| 1st Millennium CE             | 1                        | 999                    |
| 2nd Millennium CE             | 1000                     | 1999                   |
| 3rd Millennium CE             | 2000                     | 2999                   |

## 1.2 Radiocarbon age determinations (bp)

Scope Note: Absolute dates such as radiocarbon determinations form part of a very small percentage of the start and end dates of the Chronology Resources as they relate to individual finds. However, in some cases, smaller regional sequences are developed based on a range of radiocarbon dates collected at various sites. Where such absolute dates have been used, this is annotated in the 'Editorial Note' field of the Chronology Resource Model.

When dates are reported as radiocarbon ages they can be converted to calendar years through a process called calibration. This requires the original radiocarbon age determination, standard deviation, laboratory number, and material that was dated as well as the stratigraphic context(example below):

| <i>Radiocarbon age + standard deviation</i> | <i>Laboratory number</i> | <i>Sample/material dated</i> | <i>Stratigraphic Context</i> |
|---------------------------------------------|--------------------------|------------------------------|------------------------------|
| 4650±30 BP                                  | Beta-54351               | Guano sample                 | Layer B, sq. 5, spit 4.      |

(Source:<https://www.cambridge.org/core/journals/radiocarbon/article/new-radiocarbon-dates-from-archaeological-sites-in-parts-of-igboland-southeastern-nigeria/799789DA6AAE0A97FE0DE750EB0BE6A6>)

## 1.3 Protocols for radiocarbon calibration

To calibrate radiocarbon bp ages to calendrical years, the following protocols should be taken into account:

- The geographic location of where the dated sample derives. This will dictate whether to use **IntCal20** or **SHCal20**.
- The organic sample from which the radiocarbon age determination derives. The sample may derive from organic matter of terrestrial or marine origin. Each has a different **radiocarbon reservoir effect** that needs to be corrected when calibrating.
- The probability of date ranges should be reported at **2-sigma** which means that approximately **95.4%** of the measurement will fall in that range.
- Calibrated date ranges should be chosen based on their **highest percentage probability range**.
- The **earliest date** of the single range can be taken to mark the **start date**.
- Where radiocarbon dates have undergone **bayesian analysis** such dates should be reported rather than the original radiocarbon dates.

### a. *Internationally ratified radiocarbon calibration curves as of 2020*

Northern Hemisphere - **IntCal20**



Southern Hemisphere - **SHCal20**

*Note: The current extent of the calibration curve reaches as far back as 55 000 years bp for both IntCal20 and SHCal20.*

**b. Marine Reservoir Effect**

Radiocarbon ages of samples formed in the ocean, such as shells, fish, marine mammals etc., are generally several hundred years older than their terrestrial counterparts. This apparent age difference is due to the large carbon reservoir of the oceans. A correction is necessary in order to compare marine and terrestrial samples, but because of complexities in ocean circulation the actual correction varies with location. This regional difference from the average global marine reservoir correction is designated  $\Delta R$  (Stuiver and Braziunas, 1993). As a first approximation,  $\Delta R$  is assumed to be a constant for a given region and is calculated from the difference in  $^{14}\text{C}$  years of known age marine samples and the marine model age for that calendar age. (Extracted from: <http://calib.org/marine/marexpl.html>)

(Source: Reimer PJ, Reimer RW, 2001. A marine reservoir correction database and on-line interface. Radiocarbon 43:461-3).

**c. Probability: 1 or 2 sigma**

1 sigma is equal to 68% probability that a date falls within range.

2 sigma is equal to 95.4% probability that a date falls within range.

Note that calibration of radiocarbon dates in the MAEASaM chronology uses a probability of 95.4% (2 sigma).

**d. Calendar year date ranges**

Once a radiocarbon age determination has been calibrated you will receive a probability age range. For example:

Probability range at 2 sigma (95.4% probability):

338 (0.5%) 327 cal. BCE

199 cal BCE (92.8%) 255 cal. CE

285 cal CE (2.2%) 325 ca. CE

To convert to EDTF take the the first date and the last date and convert as follows:

**338 BCE - 325 CE >> -338/325**

- ORAU Oxcal Online. Link: <https://c14.arch.ox.ac.uk/oxcal.html>
- Calib I4C Radiocarbon Calibration **Execute Version 8.2.html** (M. Stuiver, P.J. Reimer, and R. Reimer). Link: <http://calib.org/calib/> *Note: Calib Version 8.1 can be downloaded as a desktop application.*